

1.

Tutorial No. 1

Page No. 2
Date

Unit No. 1: Matrices

Q.1 Do as directed

1. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix} =$

$\Rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix} =$

$= \begin{bmatrix} 1+1 & -1+2+(-1) \\ 3+1 & 4+3 \end{bmatrix}$

$= \begin{bmatrix} 2 & 0 \\ 4 & 7 \end{bmatrix}$

(a)

2. Order of $\begin{bmatrix} 3 & 9 & 0 \\ 1 & -2 & 0 \end{bmatrix}$

$\rightarrow$ (c) 2×3

3. $\begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} \times \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix} =$

$\rightarrow \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} \times \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix}$

$= \begin{bmatrix} (1)(5) + (2)(2) & (1)(6) + (2)(1) \\ (5)(5) + (6)(2) & (5)(6) + (6)(1) \end{bmatrix}$

$$= \begin{bmatrix} 5+4 & 6+2 \\ 25+12 & 30+6 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 8 \\ 37 & 36 \end{bmatrix}$$

(b)

4. If $A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $B = [3 \ 4]$ then $A \times B =$

→ જાણી,

$$A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2 \times 1} \quad B = [3 \ 4]_{1 \times 2}$$

જાણી A ની કોલમની સંખ્યા = B ની રોડની સંખ્યા
પ્રમાણે ગુણાકાર શક્ય છે. જાણી 2×2 મિટ્રિક્સ મળે.

$$\therefore A \cdot B = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [3 \ 4]$$

$$= \begin{bmatrix} (1)(3) & (1)(4) \\ (2)(3) & (2)(4) \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

(c)

$$\begin{bmatrix} (1)(1) + (2)(2) & (1)(2) + (2)(3) \\ (1)(2) + (2)(3) & (1)(3) + (2)(4) \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ 8 & 11 \end{bmatrix}$$

5.
$$\begin{bmatrix} 1 & 3 & 4 & 9 \\ 2 & 3 & 0 & 4 \\ 5 & 6 & 8 & 2 \end{bmatrix}^T =$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 5 \\ 3 & 3 & 6 \\ 4 & 0 & 8 \\ 9 & 4 & 2 \end{bmatrix}$$

(b)

6. If $\begin{bmatrix} x-3 & 2 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 4 & 0 \end{bmatrix}$ then $x =$

$$\rightarrow \textcircled{b) \ x=2} \quad x-3=5$$

$$\therefore x = 5+3$$

$$\therefore x = 8$$

(d)

7. If A is of order 3×2 and B is of order 2×1 then order of matrix AB is _____

$$\rightarrow (C) \ 3 \times 1$$

8. If A is a square matrix then all the diagonal elements of $A - A^T$ are _____

$$\rightarrow (C) \ 0$$

9. If A is of order 3×4 and B is of order 4×2 then no. of elements in AB is _____

$$\rightarrow (a) \ 6$$

10. If for a square matrix A , $A = -A^T$ then A is called _____ Matrix.
 →

11. For any matrix A , $(A^T)^T =$ _____
 → (d) A

12. If $\begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} x-2 & 2 \\ 0 & y+1 \end{bmatrix}$, then $x =$ _____

Ans $y =$ _____
 →

$$1 = x - 2$$

$$\therefore x = 1 + 2$$

$$\therefore x = 3$$

Ans

$$-3 = y + 1$$

$$\therefore y = -3 - 1$$

$$\therefore y = -4$$

(c) 3 & -4

13. If $X + \begin{bmatrix} 0 & -4 \\ 3 & 2 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ -3 & 2 \\ -5 & 4 \end{bmatrix}$ then $X =$ _____

→ Ans,

$$X + \begin{bmatrix} 0 & -4 \\ 3 & 2 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ -3 & 2 \\ -5 & 4 \end{bmatrix}$$

$$\therefore X = \begin{bmatrix} 1 & 5 \\ -3 & 2 \\ -5 & -4 \end{bmatrix} - \begin{bmatrix} 0 & -4 \\ 3 & -2 \\ 4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1-0 & 5-(-4) \\ -3-3 & 2-(-2) \\ -5-4 & -4-3 \end{bmatrix}$$

$$\therefore X = \begin{bmatrix} 1 & 9 \\ -6 & 4 \\ -9 & -7 \end{bmatrix}$$

(6)

14. A matrix A is called a λ -matrix if $\det(A)$

$$= 0.$$

→

15. If $\begin{bmatrix} 2x-3 & x-5 \\ -3 & 5 \end{bmatrix}$ is symmetric matrixes

then $x =$ _____.

$$2x-3 = x-5 \quad \text{or} \quad 2x-3 = 5$$

$$(2x-3) = (x-5) \quad \text{or} \quad 2x-3 = 5$$

$$2x-3 = 5 \quad \text{or} \quad 2x-3 = 5$$

$$2x-3 = 5 \quad \text{or} \quad 2x-3 = 5$$

$$2x-3 = 5 \quad \text{or} \quad 2x-3 = 5$$

$$2x-3 = 5 \quad \text{or} \quad 2x-3 = 5$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

Q.2 Do as Directed (3 or 4 marks)

1. Let $A = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1 & 0 \\ 7 & 8 & 9 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 & 0 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & 0 & 5 \\ 6 & 9 & -1 \\ 7 & 8 & -2 \end{bmatrix}$

Find $2A - 4B + C$.

→ Ans.

Solⁿ,
 $A = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1 & 0 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3}$, $B = \begin{bmatrix} -1 & -2 & 0 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix}_{3 \times 3}$ & $C = \begin{bmatrix} 3 & 0 & 5 \\ 6 & 9 & -1 \\ 7 & 8 & -2 \end{bmatrix}_{3 \times 3}$

→ $2A - 4B + C = 2 \cdot \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1 & 0 \\ 7 & 8 & 9 \end{bmatrix} - 4 \cdot \begin{bmatrix} -1 & -2 & 0 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 5 \\ 6 & 9 & -1 \\ 7 & 8 & -2 \end{bmatrix}$

$= \begin{bmatrix} 6 & 4 & 2 \\ 0 & 2 & 0 \\ 14 & 16 & 18 \end{bmatrix} - \begin{bmatrix} -4 & -8 & 0 \\ 4 & 4 & -4 \\ 8 & 8 & 8 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 5 \\ 6 & 9 & -1 \\ 7 & 8 & -2 \end{bmatrix}$

$= \begin{bmatrix} 6 - (-4) + 3 & 4 - (-8) + 0 & 2 - 0 + 5 \\ 0 - 4 + 6 & 2 - 4 + 9 & 0 - 4 + (-1) \\ 14 - 8 + 7 & 16 - 8 + 8 & 18 - 8 + (-2) \end{bmatrix}$

$= \begin{bmatrix} 6 + 4 + 3 & 4 + 8 + 0 & 2 - 0 + 5 \\ 0 - 4 + 6 & 2 - 4 + 9 & 0 + 4 - 1 \\ 14 - 8 + 7 & 16 - 8 + 8 & 18 - 8 - 2 \end{bmatrix}$

∴ $2A - 4B + C = \begin{bmatrix} 13 & 12 & 7 \\ 2 & 7 & 3 \\ 13 & 16 & 8 \end{bmatrix}$

2. If $M = \begin{bmatrix} -2 & 3 & 8 \\ 5 & -7 & 9 \\ 1 & -4 & 6 \end{bmatrix}$ and $N = \begin{bmatrix} 15 & -6 & 2 \\ 11 & 4 & 7 \\ 13 & 5 & 6 \end{bmatrix}$ then prove that

$$(M+N)^T = M^T + N^T$$

→ Ans. 21

$$\begin{matrix} (M+N)^T \\ M^T \\ N^T \end{matrix}$$

216)

$$M = \begin{bmatrix} -2 & 3 & 8 \\ 5 & -7 & 9 \\ 1 & -4 & 6 \end{bmatrix}_{3 \times 3} \text{ and } N = \begin{bmatrix} 15 & -6 & 2 \\ 11 & 4 & 7 \\ 13 & 5 & 6 \end{bmatrix}_{3 \times 3}$$

→ $M+N = \begin{bmatrix} -2 & 3 & 8 \\ 5 & -7 & 9 \\ 1 & -4 & 6 \end{bmatrix} + \begin{bmatrix} 15 & -6 & 2 \\ 11 & 4 & 7 \\ 13 & 5 & 6 \end{bmatrix}$

$$= \begin{bmatrix} -2+15 & 3+(-6) & 8+2 \\ 5+11 & -7+4 & 9+7 \\ 1+13 & -4+5 & 6+6 \end{bmatrix}$$

$$M+N = \begin{bmatrix} 13 & -3 & 10 \\ 16 & -3 & 16 \\ 14 & 1 & 12 \end{bmatrix}$$

$$\therefore (M+N)^T = \begin{bmatrix} 13 & 16 & 14 \\ -3 & -3 & 1 \\ 10 & 16 & 12 \end{bmatrix} \quad \text{--- (1)}$$

→ $M^T = \begin{bmatrix} -2 & 3 & 8 \\ 5 & -7 & 9 \\ 1 & -4 & 6 \end{bmatrix}^T = \begin{bmatrix} -2 & 5 & 1 \\ 3 & -7 & -4 \\ 8 & 9 & 6 \end{bmatrix}$

$$\rightarrow N^T = \begin{bmatrix} 15 & -6 & 2 \\ 11 & 4 & 7 \\ 13 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 15 & 11 & 13 \\ -6 & 4 & 5 \\ 2 & 7 & 6 \end{bmatrix}$$

$$\rightarrow \therefore M^T + N^T = \begin{bmatrix} -2 & 5 & 1 \\ 3 & -7 & -4 \\ 8 & 9 & 6 \end{bmatrix} + \begin{bmatrix} 15 & 11 & 13 \\ -6 & 4 & 5 \\ 2 & 7 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} -2+15 & 5+11 & 1+13 \\ 3+(-6) & -7+4 & -4+5 \\ 8+2 & 9+7 & 6+6 \end{bmatrix}$$

$$\therefore M^T + N^T = \begin{bmatrix} 13 & 16 & 14 \\ -3 & -3 & 1 \\ 10 & 16 & 12 \end{bmatrix} \quad \text{--- (2)}$$

$\rightarrow$ $\therefore$ એમી. 1 અને એમી. 2 ની સરખામણી સાબીત થાય છે કે $(M+N)^T = M^T + N^T$ છે.

3. $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ & $B = \begin{bmatrix} -2 & 3 \\ 1 & 1 \end{bmatrix}$ then prove that, $A^T B^T = (BA)^T$

$\rightarrow$ Ans.

જાણી,

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \quad \& \quad B = \begin{bmatrix} -2 & 3 \\ 1 & 1 \end{bmatrix}$$

$$\rightarrow A^T = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \& \quad B^T = \begin{bmatrix} -2 & 1 \\ 3 & 1 \end{bmatrix}$$

→ ex, $A^T \cdot B^T = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} -2 & 1 \\ 3 & 1 \end{bmatrix}$

$$= \begin{bmatrix} (1)(-2) + (2)(3) & (1)(1) + (2)(1) \\ (3)(-2) + (4)(3) & (3)(1) + (4)(1) \end{bmatrix}$$

$$= \begin{bmatrix} -2 + 6 & 1 + 2 \\ -6 + 12 & 3 + 4 \end{bmatrix}$$

$$\therefore A^T B^T = \begin{bmatrix} 4 & 3 \\ 6 & 7 \end{bmatrix} \quad \text{--- (I)}$$

→ ex,

$$B = \begin{bmatrix} -2 & 3 \\ 1 & 1 \end{bmatrix} \quad \& \quad A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

ex,

$$(B \cdot A) = \begin{bmatrix} -2 & 3 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} (-2)(1) + (3)(2) & (-2)(3) + (3)(4) \\ (1)(1) + (1)(2) & (1)(3) + (1)(4) \end{bmatrix}$$

$$= \begin{bmatrix} -2 + 6 & -6 + 12 \\ 1 + 2 & 3 + 4 \end{bmatrix}$$

$$(BA) = \begin{bmatrix} 4 & 6 \\ 3 & 7 \end{bmatrix}$$

$$\therefore (BA)^T = \begin{bmatrix} 4 & 3 \\ 6 & 7 \end{bmatrix} \quad \text{--- (2)}$$

→ ∴ સમી. 1 અને સમી. 2 ની સમાનતામાં આપો
પરિણતર $A^T \cdot B^T = (BA)^T$ છે.

4. If $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ then find $A + A^T + I$.

→ Ans.

જાણી,

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$$

→ જા. જા. છે. કે. A નો $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

→ છે,

$$A + A^T + I = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+1+1 & (-1)+2+0 \\ 2+(-1)+0 & 3+3+1 \end{bmatrix}$$

$$\rightarrow \therefore A + A^T + I = \begin{bmatrix} 3 & 1 \\ 1 & 7 \end{bmatrix}$$

5. If $\begin{bmatrix} x+3 & -6 & 2 \\ y+1 & 2 & 0 \\ z-3 & -21 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -6 & 2 \\ -3 & 2 & 0 \\ 4x & 21 & 0 \end{bmatrix}$ then find the

Values of x, y & z .

$\rightarrow$ સાલ,

દાતી કે

$$A = \begin{bmatrix} x+3 & -6 & 2 \\ y+1 & 2 & 0 \\ z-3 & -21 & 0 \end{bmatrix} \quad \& \quad B = \begin{bmatrix} 0 & -6 & 2 \\ -3 & 2 & 0 \\ 4x & 21 & 0 \end{bmatrix}$$

$\rightarrow$ A અને B ની મેમ્બર્સ

$$\therefore x+3 = 0$$

$$\therefore x = -3$$

$$\therefore y+1 = -3$$

$$\therefore y = -2$$

$$\therefore z-3 = 4x$$

$$\therefore z-3 = 4(-3)$$

$$\therefore z = -12+3$$

$$\therefore z = -9$$

6. If $A = \begin{bmatrix} -1 & 2 \\ -2 & 4 \end{bmatrix}$ then find $A^2 + I$ $\therefore A + A$ $\therefore$ $\leftarrow$

$\rightarrow$ Ans.

જાણી,

$$A = \begin{bmatrix} -1 & 2 \\ -2 & 4 \end{bmatrix}_{2 \times 2}$$

$$A^2 = A \cdot A$$

$$= \begin{bmatrix} -1 & 2 \\ -2 & 4 \end{bmatrix} \cdot \begin{bmatrix} -1 & 2 \\ -2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} (-1)(-1) + (2)(-2) & (-1)(2) + (2)(4) \\ (-2)(-1) + (4)(-2) & (-2)(2) + (4)(4) \end{bmatrix}$$

$$= \begin{bmatrix} 1 + (-4) & -2 + 8 \\ 2 + (-8) & -4 + 16 \end{bmatrix}$$

$$\cancel{A^2} = \begin{bmatrix} 5 & 6 \\ 10 & 12 \end{bmatrix} \quad A^2 = \begin{bmatrix} -3 & 6 \\ -6 & 12 \end{bmatrix}$$

જા. જા. જા. ૩

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\rightarrow \therefore \cancel{A^2} + I = \begin{bmatrix} 5 & 6 \\ 10 & 12 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5+1 & 6+0 \\ 10+0 & 12+1 \end{bmatrix}$$

$$A^2 + I = \begin{bmatrix} 6 & 6 \\ 10 & 13 \end{bmatrix}$$

$$\rightarrow \therefore A^2 + I = \begin{bmatrix} -3 & 6 \\ -6 & 12 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3+1 & 6+0 \\ -6+0 & 12+1 \end{bmatrix}$$

$$\therefore A^2 + I = \begin{bmatrix} -2 & 6 \\ -6 & 13 \end{bmatrix}$$

7. Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 3 & 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 2 & 1 \\ 1 & 7 & 5 \end{bmatrix}$, Find AB .

$\rightarrow$ Ans. જાણી,

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 3 & 2 & 1 \end{bmatrix}_{2 \times 3} \quad \& \quad B = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 2 & 1 \\ 1 & 7 & 5 \end{bmatrix}_{3 \times 3}$$

જાણી,

$$A \cdot B =$$

$\rightarrow$ જવાબ,

A ના કોલમની સંખ્યા = B ની રોની સંખ્યા

તેથી AB જાણ્યું છે અને તે 2×3 સ્કેલર વાળી સંરેખ વાળી.

$$\therefore AB = \begin{bmatrix} 1 & -1 & 1 \\ 3 & 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 1 \\ 4 & 2 & 1 \\ 1 & 7 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(1) + (-1)(4) + (1)(1) & (1)(2) + (-1)(2) + (1)(7) \\ (3)(1) + (2)(4) + (1)(1) & (3)(2) + (2)(2) + (1)(7) \end{bmatrix}$$

$$(1)(1) + (-1)(4) + (1)(1)$$

$$(3)(1) + (2)(4) + (1)(1)$$

$$= \begin{bmatrix} 1 + (-4) + 1 & 2 + (-2) + 7 & 1 + (-1) + 5 \\ 3 + 8 + 1 & 6 + 4 + 7 & 3 + 2 + 5 \end{bmatrix}$$

→

$$\therefore AB = \begin{bmatrix} -2 & 7 & 5 \\ 12 & 17 & 10 \end{bmatrix}_{2 \times 3}$$

8. If $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & 0 \\ -1 & 4 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$

then, Prove that $AB = AC$.

→

Ans.

જાણી,

બંને AB ની બાબત

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & 0 \\ -1 & 4 & 0 \end{bmatrix}_{3 \times 3} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix}_{3 \times 3}$$

→

Since,

A ની કોલમની સંખ્યા = B ની રો ની સંખ્યા

તેથી AB શક્ય છે. અને તે 3×3 આંકવાળી કિટ્ટી બની.

$$\therefore AB = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & 0 \\ -1 & 4 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(1) + (2)(1) + (0)(2) & (1)(2) + (2)(1) + (0)(2) \\ (1)(1) + (1)(1) + (0)(2) & (1)(2) + (1)(1) + (0)(2) \\ (-1)(1) + (4)(1) + (0)(2) & (-1)(2) + (4)(1) + (0)(2) \end{bmatrix}$$

$$\begin{bmatrix} (1)(3) + (2)(-1) + (0)(2) \\ (1)(3) + (1)(-1) + (0)(2) \\ (-1)(3) + (4)(-1) + (0)(2) \end{bmatrix}$$

$$= \begin{bmatrix} 1+2+0 & 2+2+0 & 3+(-2)+0 \\ 1+1+0 & 2+1+0 & 3+(-1)+0 \\ -1+4+0 & -2+4+0 & -3+(-4)+0 \end{bmatrix}$$

$$\rightarrow \therefore AB = \begin{bmatrix} 3 & 4 & 1 \\ 2 & 3 & 2 \\ 3 & 2 & -7 \end{bmatrix} \quad \text{--- (1)}$$

$$\rightarrow \text{AC મિલેલ}$$

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & 0 \\ -1 & 4 & 0 \end{bmatrix}_{3 \times 3} \quad \& \quad C = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}_{3 \times 3}$$

Since,

A ના કોલમની સંખ્યા = C ની રોની સંખ્યા

તેથી AB ગણ્ય છે અને તે 3×3 આઈડેન્ટિટી મેટ્રિક્સ થશે.

$$\therefore AC = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & 0 \\ -1 & 4 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} =$$

$$= \begin{bmatrix} (1)(1) + (2)(1) + (0)(1) & (1)(2) + (2)(1) + (0)(1) & (1)(3) + (2)(-1) + (0)(1) \\ (1)(1) + (1)(1) + (0)(1) & (1)(2) + (1)(1) + (0)(1) & (1)(3) + (1)(-1) + (0)(1) \\ (-1)(1) + (4)(1) + (0)(1) & (-1)(2) + (4)(1) + (0)(1) & (-1)(3) + (4)(-1) + (0)(1) \end{bmatrix}$$

$$= \begin{bmatrix} 1+2+0 & 2+2+0 & 3+(-2)+0 \\ 1+1+0 & 2+1+0 & 3+(-1)+0 \\ -1+4+0 & -2+4+0 & -3+(-4)+0 \end{bmatrix}$$

$$\therefore AC = \begin{bmatrix} 3 & 4 & 1 \\ 2 & 3 & 2 \\ 3 & 2 & -7 \end{bmatrix} \quad \text{--- (2)}$$

જો સમી. 1 અને સમી. 2 નો અનુસરણ કરીએ તો
આબોધ થાય છે કે $AB = AC$ થાય.

9. For $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}$ find AB and BA

Whichever is possible.

→ Ans.

જો સમી. 1 નો અનુસરણ કરીએ તો
પ્રથમ AB શક્ય

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \end{bmatrix}_{2 \times 3} \quad B = \begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}_{2 \times 2}$$

→ Since,

A ના કોલમોની સંખ્યા $\neq B$ ના રોનો સંખ્યા
તેથી AB શક્ય નથી.

→ BA શું થાય છે

$$B = \begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}_{2 \times 2} \quad \& \quad A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \end{bmatrix}_{2 \times 3}$$

→ Since,

B ની કોલમોની સંખ્યા = A ની રોનો સંખ્યા
કેથી BA શક્ય છે અને તે 2×3 સંખ્યા
વાળો ફરેક મળે

$$\therefore BA = \begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} (0)(1) + (1)(2) & (0)(-1) + (1)(3) & (0)(2) + (1)(5) \\ (4)(1) + (2)(2) & (4)(-1) + (2)(3) & (4)(2) + (2)(5) \end{bmatrix}$$

$$= \begin{bmatrix} 0+2 & 0+3 & 0+5 \\ 4+4 & -4+6 & 8+10 \end{bmatrix}$$

→

$$\therefore BA = \begin{bmatrix} 2 & 3 & 5 \\ 8 & 2 & 18 \end{bmatrix}_{2 \times 3}$$

10. If $A+B = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix}$ and $A-B = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$ then Find AB.

→ Ans.

અમો, જાણી જો

$$A = ?$$

$$B = ?$$

$$AB = ?$$

$$A+B = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix} \quad \text{--- (I)}$$

$$A-B = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} \quad \text{--- (2)}$$

→

અમી. 1 અને અમી. 2 ની સરવાળો કરતા

$$\therefore (A+B) + (A-B) = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$$

$$\therefore A+B+A-B = \begin{bmatrix} 1+3 & -1+1 \\ 3+1 & 0+4 \end{bmatrix}$$

$$\therefore A+A = \begin{bmatrix} 4 & 0 \\ 4 & 4 \end{bmatrix}$$

$$\therefore 2A = \begin{bmatrix} 4 & 0 \\ 4 & 4 \end{bmatrix}$$

$$\therefore A = \frac{1}{2} \begin{bmatrix} 4 & 0 \\ 4 & 4 \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 4/2 & 0/2 \\ 4/2 & 4/2 \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}$$

→

A ની કિંમત ને અમી. 1 માં મુકતા.

$$A+B = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix} + B = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1-2 & -1-0 \\ 3-2 & 0-2 \end{bmatrix}$$

$$\therefore B = \begin{bmatrix} -1 & -1 \\ 1 & -2 \end{bmatrix}$$

→ હો,

$$A = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}_{2 \times 2} \quad \& \quad B = \begin{bmatrix} -1 & -1 \\ 1 & -2 \end{bmatrix}_{2 \times 2}$$

Since,

A ની કોણકોની માત્રા = B ની કોણકોની માત્રા
તેથી AB બાકર હો જાકી સે 2x2 બાકર નો
નિરૂપ માળગો.

$$\therefore AB = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} -1 & -1 \\ 1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} (2)(-1) + (0)(1) & (2)(-1) + (0)(-2) \\ (2)(-1) + (2)(1) & (2)(-1) + (2)(-2) \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} -2+0 & -2+0 \\ -2+2 & -2+(-4) \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} -2 & -2 \\ 0 & -6 \end{bmatrix}$$